

VECTORS:

1. An ordered n -tuple of numbers is called an n -vector. If $x_1, x_2 \dots x_n$ be any n numbers, then the ordered n -tuple

$x = (x_1, x_2 \dots x_n)$ is called an n -vector.

The n numbers $x_1, x_2 \dots x_n$ are called components of the n -vector x .

2. A vector may be written as either a vector or a column vector.

If A be a matrix of order $m \times n$, then each column of A will be a m -vector and each row of A will be an n -vector.

3. A vector whose components are all zero is called zero vector and will be denoted by $\mathbf{0}$.

4. A vector over real numbers is called a real vector and the vector over complex numbers is called a complex vector.

eg: $[4 \ 2 \ 0 \ 6]$, $\begin{bmatrix} 2 \\ 4 \\ 0 \\ 6 \end{bmatrix}$ are row and column vectors respectively.

Linear Dependence And Linear Independence of vectors:

1. Linearly dependent set of vectors:

A set of ' r ' n -vectors $x_1, x_2, \dots, x_r$ is said to be linearly dependent

if there exists r scalars $k_1, k_2, \dots, k_r$ not all zero, such that

$$k_1 x_1 + k_2 x_2 + \dots + k_r x_r = 0,$$

where ' 0 ' denotes the n -vector whose components are all zero.

2. Linearly independent set of vectors:

A set of ' r ' n -vectors $x_1, x_2, \dots, x_r$ is said to be linearly independent,

if every relation of the type

$$k_1 x_1 + k_2 x_2 + \dots + k_r x_r = 0$$

$$\Rightarrow k_1 = k_2 = \dots = k_r = 0$$

where $k_1, k_2, \dots, k_r$ are scalars.

3. A vector as a linear combination of vectors:

A vector X which can be expressed in the form

$X = k_1 x_1 + k_2 x_2 \dots + k_r x_r$ is said to be a linear combination of the set of vectors $x_1, x_2 \dots x_r$.

Here $k_1, k_2 \dots k_r$ are any scalars.

Now the following two results are quite obvious:

- (i) If a set of vectors is linearly dependent, then at least one member of the set can be expressed as a linear combination of the remaining members.
- (ii) If a set of vectors is linearly independent then no member of the set can be expressed as the linear combination of the remaining members.

$\text{---} x \text{---}$

* Show that vectors $x_1 = (1, 2, 3)$, $x_2 = (3, 6, 12)$ are linearly dependent.

sol: By taking $k_1 = 3$, $k_2 = -1$

$$\begin{aligned}k_1 x_1 + k_2 x_2 &= 3x_1 + (-1)x_2 \\&= 3(1, 2, 3) + (-1)(3, 6, 12) \\&= (3, 6, 12) + (-3, -6, -12) \\&= (3-3, 6-6, 12-12) \\&= (0, 0, 0)\end{aligned}$$

Thus there exists two scalars $k_1 = 3$, $k_2 = -1$, which are not all zero s.t

$$k_1 x_1 + k_2 x_2 = 0$$

Hence the vectors x_1, x_2 are linearly dependent.

* Show that the vectors $x_1 = (1, 0, 0)$, $x_2 = (0, 1, 0)$, $x_3 = (0, 0, 1)$ are linearly independent.

sol. Given $x_1 = (1, 0, 0)$,

$$x_2 = (0, 1, 0)$$

$$x_3 = (0, 0, 1) \text{ are three vectors.}$$

Let k_1, k_2, k_3 be any three scalars satisfying the relation

$$k_1 x_1 + k_2 x_2 + k_3 x_3 = 0$$

$$\text{i.e. } k_1 (1, 0, 0) + k_2 (0, 1, 0) + k_3 (0, 0, 1) = (0, 0, 0)$$

$$\text{i.e. } (k_1, 0, 0) + (0, k_2, 0) + (0, 0, k_3) = (0, 0, 0)$$

$$\text{i.e. } (k_1, k_2, k_3) = (0, 0, 0)$$

Obviously this relation is true ($\Rightarrow$)

$$\text{if } k_1 = 0, k_2 = 0, k_3 = 0.$$

Hence the vectors x_1, x_2 and x_3 are linearly independent.

*. Show that the set consisting only of the zero vector is linearly dependent.

Sol. Let $x = (0, 0, \dots, 0)$ be an n -vector, whose components are all zero.

then the relation $kx = 0$ is true for some non-zero scalar k i.e. $k \neq 0$.

for example:

$$1x = 0 \quad \text{where } 1 \neq 0$$

Hence the zero vector ' x ' is linearly dependent.

$$= x =$$

* Show that the vectors $x_1 = (1, 2, 3)$, $x_2 = (2, -2, 0)$ form a linearly independent set.

Sol: Consider the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & -2 & 0 \end{bmatrix}_{2 \times 3}$

It has 2-rowed and 1-rowed minors

Of these there is at least one 2-rowed minor

$$\text{i.e. } \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix} = -2 - 4 = -6 \neq 0.$$

$\therefore$ The rank of the matrix A is 2 and this rank indicates the maximum number of linearly independent rows of A .

Hence the vectors $(1, 2, 3)$ and $(2, -2, 0)$ are linearly independent.

$= \times =$

* Show that the vectors $x_1 = (2, -2, 1)$, $x_2 = (1, 4, -1)$ and $x_3 = (4, 6, -3)$ are linearly independent.

Sol. Method:

Given x_1, x_2, x_3 are the vectors.

Let k_1, k_2, k_3 scalars satisfying the relation

$$k_1 x_1 + k_2 x_2 + k_3 x_3 = 0$$

$$k_1 (2, -2, 1) + k_2 (1, 4, -1) + k_3 (4, 6, -3) = 0$$

$$(2k_1, -2k_1, k_1) + (k_2, 4k_2, -k_2) + (4k_3, 6k_3, -3k_3) = (0, 0, 0)$$

$$2k_1 + k_2 + 4k_3 = 0 \quad \text{--- (1)}$$

$$-2k_1 + 4k_2 + 6k_3 = 0 \quad \text{--- (2)}$$

$$k_1 - k_2 - 3k_3 = 0 \quad \text{--- (3)}$$

Solving (1), (2) and (3), we get $k_1 = k_2 = k_3 = 0$

Hence the given vectors x_1, x_2, x_3 are linearly independent.

Method: Here the vectors x_1, x_2, x_3 form a square matrix

$$\text{Let } A = \begin{bmatrix} 2 & -2 & 1 \\ 1 & 4 & -1 \\ 4 & 6 & -3 \end{bmatrix}$$

$$R_1 \leftrightarrow R_2$$

$$\sim \begin{bmatrix} 1 & 4 & -1 \\ 2 & -2 & 1 \\ 4 & 6 & -3 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1, \quad R_3 \rightarrow R_3 - 4R_1$$

$$\sim \begin{bmatrix} 1 & 4 & -1 \\ 0 & -10 & 3 \\ 0 & -10 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\sim \begin{bmatrix} 1 & 4 & -1 \\ 0 & -10 & 3 \\ 0 & 0 & -2 \end{bmatrix}$$

which is in Echelon form.

Hence rank of A = the no. of non-zeros of A = 3
and this indicate the maximum number of linearly independent rows of A .

Hence the given vectors x_1, x_2, x_3 are L.I.

$$= x =$$

* Show that the vectors

$x_1 = (1, 1, 2)$, $x_2 = (1, 2, 5)$, $x_3 = (5, 3, 4)$
are linearly dependent. Also express vector
as a linear combination.

* Show that the vectors are linearly dependent.
Also express each vectors as a linear combination

a) $(3, 2, 7)$, $(2, 4, 1)$, $(1, -2, 6)$

b) $(1, 3, 4, 2)$, $(3, -5, 2, 2)$, $(2, -1, 3, 2)$

* Prove that the following vectors are linearly independent.

a) $(1, 1, 1)$; $(3, 1, 2)$; $(2, 1, 4)$

b) $(1, 1, 1, 3)$; $(1, 2, 3, 4)$; $(2, 3, 4, 9)$

* Find the values of a and b so that the rank of $\begin{bmatrix} 1 & -2 & 3 & -1 \\ 2 & 1 & -1 & 2 \\ 0 & -2 & a & b \end{bmatrix}$ is 3.